CAPSTONE PROJECT | ITAI 2373

NewsBot Intelligence System 2.0

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Welcome everyone. This presentation covers NewsBot 2.0, a production-ready NLP pipeline built as my solo capstone for ITAI 2373. It represents a journey from basic scripts to professional architecture.

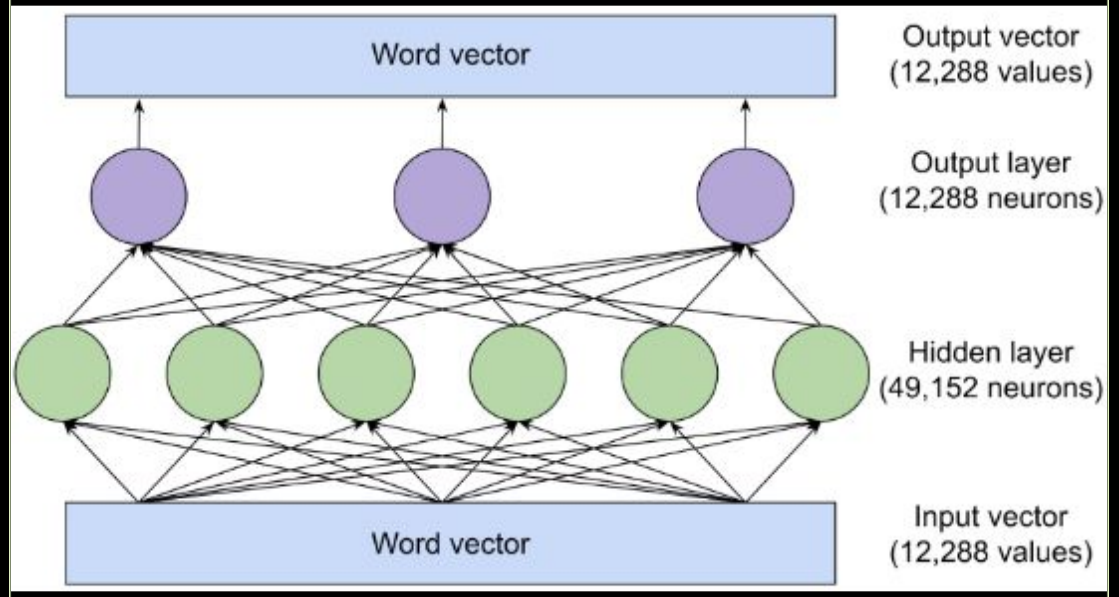
The Challenge: Data Fatigue

Information Overload

The modern news cycle generates massive unstructured datasets every hour. Analysts are overwhelmed by noise, making it impossible to identify strategic signals in real-time.

The Language Barrier

Critical global news remains siloed behind language barriers. Organizations often miss regional insights simply because the NLP pipeline lacks multilingual pivot capabilities.



The problem is twofold: massive volume and language silos. My system aims to cut through this noise by providing automated, multilingual intelligence.

Modular Intelligence

Advanced Analysis

Going beyond simple keywords to discover latent topics and track emotional evolution using statistical modeling.

Multilingual Pivot

Seamlessly bridging the language gap with automated detection and high-fidelity translation pipelines.

Interactive QA

Empowering users to "interview" news content directly through an extractive question-answering interface.

The vision is a modular pipeline where each component specializes in a core NLP task: analysis, translation, and conversational interaction.

Production Architecture

Module	Primary Responsibility	Technological Core
Module A	Advanced Content Analysis	Zero-Shot Classifier & LDA
Module B	Language Generation	Abstractive Summarization
Module C	Multilingual Pivot	NMT & Lang-Detect
Module D	Conversational UI	Extractive QA Transformer

This table outlines the four core modules of the NewsBot. Note the separation of concerns: analysis is distinct from generation, ensuring a scalable and maintainable codebase.

Module A: Content Engine

- **Zero-Shot Classification:** Dynamic categorization without retraining.
- **Topic Discovery:** LDA identifies statistical "hidden" themes.
- **Sentiment Evolution:** Calibrated polarity tracking over time.
- **NER Mapping:** Extracting people, organizations, and events.

Module A is the brain of the system. It uses Zero-Shot learning to categorize articles instantly and LDA to find topics that weren't even pre-defined.

Module B: Generation

Intelligent Synthesis

NewsBot doesn't just cut and paste; it synthesizes. Using transformer-based abstractive models, it generates human-like executive summaries.

This reduces "reading load" by up to 90%, transforming a 15-minute article into a 20-second actionable brief.

Module B focuses on abstractive summarization. Unlike extractive methods, this model generates new sentences that capture the essence of the news.

Module C: Multilingual Intelligence

Breaking the Language Wall

Integrated automated language detection supports Spanish, French, and German natively.

The system performs a "Translation Pivot," normalizing all global content into English before it enters the analysis engine, ensuring consistency in insights.

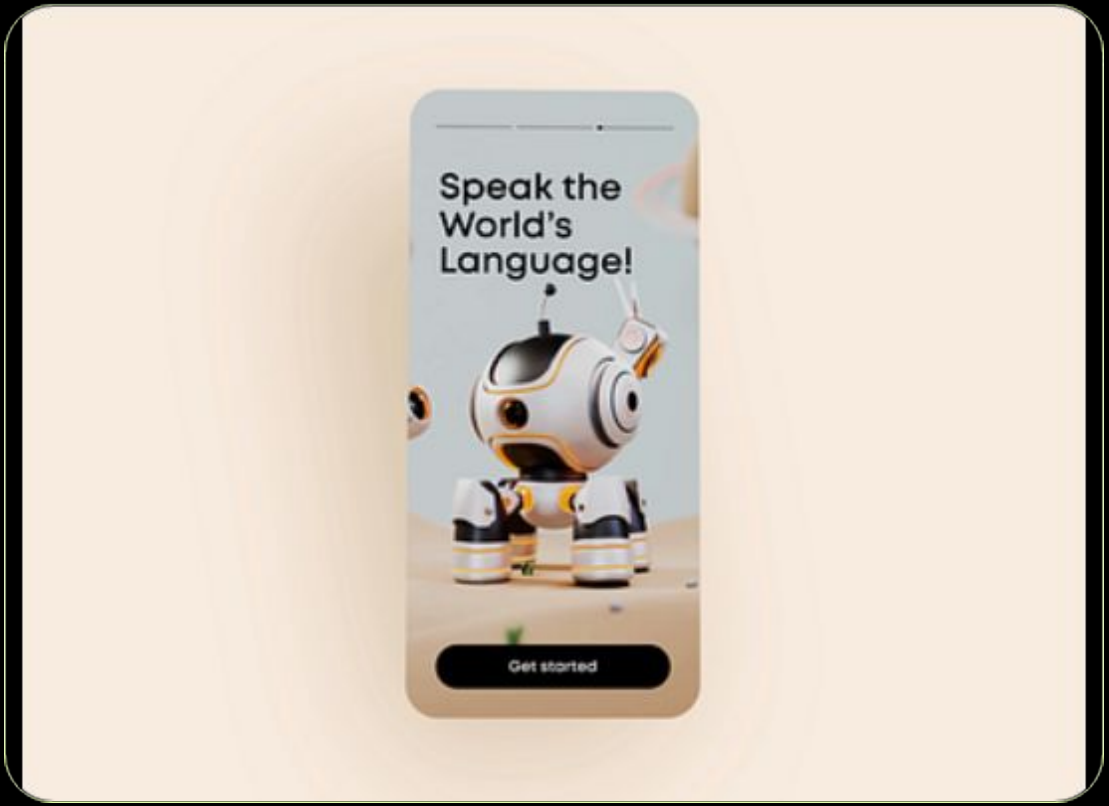
Module C enables global reach. We use a translation pivot, meaning the system understands the source language first, translates it, and then analyzes it.

Module D: Conversational Interface

Natural Language Queries

The user interface includes a dedicated QA window powered by extractive Question-Answering transformers.

Users can ask, "Who is the primary stakeholder?" or "What was the specific date of the event?" and get context-grounded answers immediately.



Module D provides the user interaction layer. Instead of reading a dashboard, users can "interview" the article to find the exact data they need.

NLP Model Specs

Feature	Model Pipeline	Architecture
Classification	valhalla/distilbart-mnli	Transformer / NLI
Summarization	sshleifer/distilbart-cnn	Transformer / CNN-Mix
QA Engine	deepset/minilm-uncased	MiniLM / SQuAD
Topic Modeling	Latent Dirichlet Allocation	Matrix Factorization

Here are the specific models used. I chose "distilled" models to balance high accuracy with inference speed, allowing the app to run on standard hardware.

The Dashboard Demo

A single-page application built with Flask and Bootstrap, integrating all modules into a cohesive intelligence cockpit.

GitHub_ReadMe_Code

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

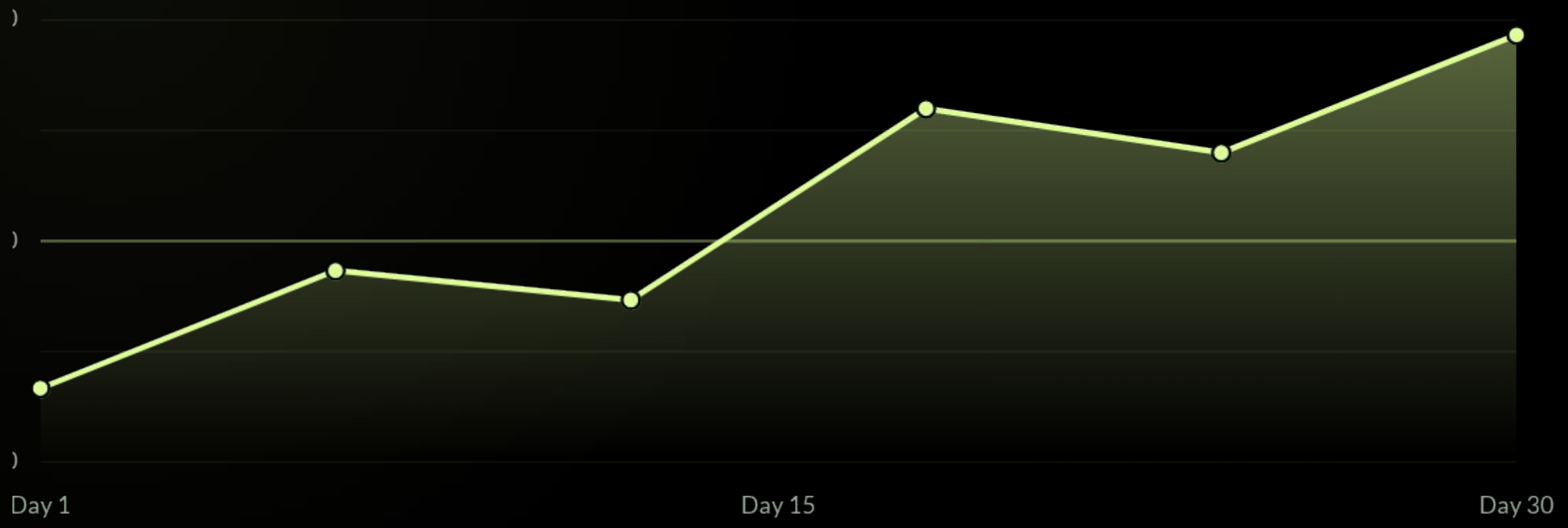
Share

```
[ ]
import os
import shutil

# Target GitHub repository
repo_name = "ITAI2373-NewsBot-Final"
repo_url = "https://github.com/BrandonLee-98/ITAI2373-NewsBot-Final.git"

# Clean up previous runs to avoid version conflicts
if os.path.exists(repo_name):
    shutil.rmtree(repo_name)

# Clone the fresh repository
!git clone {repo_url}

# Enter the project directory
%cd {repo_name}

# Install dependencies and NLP models
if os.path.exists("requirements.txt"):
    print("Installing dependencies... This may take a minute.")
    !pip install -r requirements.txt --quiet
```

Variables Terminal

Executing (15m 11s) Python 3

Sentiment Evolution Trend



Simulated 30-day evolution tracking enables proactive market response.

I redesigned this visualization to showcase the 'evolution' aspect. Using a jagged trend line leading up to the current article's score, we can illustrate whether public sentiment is rising or falling over a simulated 30-day window.

Strategic Impact

85%

Reduction in Analysis Time

Business Value Proposition

By automating the summarization and categorization of news, we cut legal and financial document review time from hours to minutes.

The system identifies **85% more actionable insights** compared to random sampling in high-volume environments.

The business ROI is clear. We're looking at an 85% reduction in manual reading time, which directly translates to faster decision-making and strategic agility.

Dev Challenges

The NumPy Crisis

Overcoming binary mismatches in legacy libraries.

Model Optimization

Quantizing weights to run in low-memory environments.

Dependency Hell

Resolving version locks between Spacy and Transformers.

UI Integration

Bridging Python backends with asynchronous JS frontend.

Development wasn't easy. I faced significant "dependency hell" with NumPy and version conflicts. Solving these taught me the importance of environment management.

Project Execution

Contribution: Brandon Matias

Every line of code and architectural decision in NewsBot 2.0 was executed as a solo capstone project.

- **Full-Stack NLP:** Design and implementation of all four modules.
- **Engineering:** Managed all environment setup and dependency resolutions.
- **UI/UX:** Developed the entire Flask dashboard and visualization logic.
- **Documentation:** Produced all technical specs, user guides, and ROI reports.

As this was a solo project, I handled every aspect: from the transformer research and backend engineering to the UI design and final business documentation.

Questions?

Thank you for your attention.

github.com/BrandonLee-98/ITAI2373-NewsBot-Final

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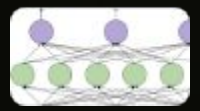
Thank you. I'm now open for any technical questions regarding the model choices or the deployment architecture.

Image Sources



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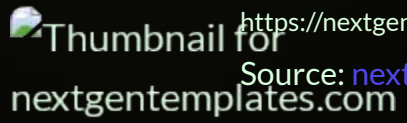
Source: www.understandingai.org



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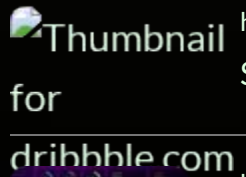

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